BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023 • Paper Code: —

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Objective Type

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Tyndall effect is shown by:

(A) Sugar solution
(B) Paints
(C) Jelly
(D) Chalk solution

Q2. Identify the compound which has covalent bond:

(A) NaCl
(B) KCl
(C) CaO
(D) HCl

Q3. One of the following is non-metal, identify it:

(A) Na
(B) Ca
(C) Br
(D) Cr

Q4. How many number of moles are equivalent to 8 gm of O_2:

(A) 0.15
(B) 0.18
(C) 0.21
(D) 0.24

Q5. The most common example of corrosion is:

(A) Chemical decay
(B) Rusting of Iron
(C) Rusting of aluminium
(D) Rusting of tin

Q6. The mass number of sodium is:

(A) 19 (B) 23

(C) 27 (D) 31

Q7. The oxidation number of hydrogen in HCl is:

(A) +1 (B) −1

(C) +2 (D) −2

Q8. The third period contains elements:

(A) 32 (B) 18

(C) 8 (D) 2

Q9. The P subshell has:

(A) One orbital (B) Two orbitals

(C) Three orbitals (D) Four orbitals

Q10. The fastest diffuses gas is:

(A) Hydrogen (B) Helium

(C) Fluorine (D) Chlorine

Q11. Single covalent bond formation involves electrons:

(A) 2 (B) 4

(C) 6 (D) 8

Q12. Air is an example of solution:

(A) Liquid in gas (B) Solid in gas

(C) Gas in gas (D) Gas in liquid

Essay Type — PART – I

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

Q2. (Marks 10) Write short answers to any Five (5) questions:

(i) Define Avogadro's Number.

(ii) What is meant by mole?

(iii) Differentiate between shell and subshell.

(iv) Write electronic configuration of argon (Ar=18)

(v) Write the trend of shielding effect in period and group.

(vi) Define electronegativity.

(vii) Why atomic number is a more fundamental property than atomic mass?

(viii) Write names of elements of first period.

Q3. (Marks 10) Write short answers to any Five (5) questions:

(i) Why do atoms react?

(ii) Define chemical bond.

(iii) Write name of different types of chemical bond.

(iv) What is diffusion? Explain with an example.

(v) Define standard atmospheric pressure. What are its units?

(vi) What is difference between dilute solution and concentrated solution?

(vii) Define unsaturated solution.

(viii) Give an example for "Gas in Solid" solution.

Q4. (Marks 10) Write short answers to any Five (5) questions:

(i) Write difference between spontaneous and non-spontaneous reactions.

(ii) Define non-electrolytes and give example.

(iii) Define corrosion.

(iv) What do you mean by galvanizing? Write its advantages.

(v) Write two physical properties of metals.

(vi) What is the trend of electropositivity in group?

(vii) Why platinum is used by jewellery making?

(viii) Write reaction of fluorine with water.

> Note: Attempt any TWO questions. (PART – II)

Q5.

(a) Calculate the number of moles and number of molecules present in 6 grams of water.

(b) Write any two results of Ruther Ford's experiment and two defects of Ruther Ford's atomic model.

Q6.

(a) What is an ionic bond? Discuss the formation of ionic bond between sodium and chlorine atom.

(b) Differentiate between crystalline and amorphous cells.

Q7.

(a) What is galvanic cell? Write four differences between electrolytic and galvanic cell.

(b) Define solubility. Describe effect of temperature on solubility.

GROUP – II

Objective Type

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Molarity is the number of moles of solute dissolved in:

(A) One kg of solution
(B) 1 dm^3 of solution
(C) 100 g of solvent
(D) 1 dm^3 of solvent

Q2. The number of moles which are equivalent to 5 grams of CO_2

(A) 0.15
(B) 0.18
(C) 0.21
(D) 0.24

Q3. Which one of the following non-metal is lustrous

(A) Iodine
(B) Carbon
(C) Phosphorus
(D) Sulphur

Q4. The compound that is a heteroatomic molecule

(A) Cl_2
(B) H_2O_2
(C) O_3
(D) H_2

Q5. Oxidation number of all elements in the free state is

(A) +1
(B) zero
(C) −1
(D) +3

Q6. The number of subshells in N shell is

(A) One
(B) Two
(C) Three
(D) Four

Q7. In the redox reaction between Zn and HCl, the oxidizing agent it

(A) Zn
(B) H_2
(C) H^+
(D) Cl^-

Q8. A bond formed between two non-metals is

(A) Covalent
(B) Coordinate covalent
(C) Ionic
(D) Metallic

Q9. The element that has highest electronegativity

(A) Chlorine
(B) Bromine
(C) Fluorine
(D) Iodine

Q10. Vapour pressure of water in mmHg at 80°C is

(A) 17.5
(B) 55.3
(C) 149.4
(D) 355.1

Q11. Identify the compound which is not soluble in water

(A) $MgCl_2$
(B) KBr
(C) NaCl
(D) C_6H_6

Q12. Which one of the following is heterogeneous mixture

(A) Salt solution
(B) Alcohol in water
(C) Milk of magnesia
(D) Sugar solution

Essay Type — PART – I

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

Q2. (Marks 10) Write short answers to any Five (5) questions:

(i) Define industrial chemistry and analytical chemistry.

(ii) Explain why hydrogen and oxygen are considered as elements whereas water as a compound?

(iii) Write two properties of neutron.

(iv) How many maximum electrons can be accommodated in K, L, M, N shells?

(v) What is the trend of ionization energy in period and why?

(vi) Define electronegativity. Which one halogen has highest electronegativity value?

(vii) Write names of elements of first period.

(viii) Write down the name of elements of group-I (any two).

Q3. (Marks 10) Write short answers to any Five (5) questions:

(i) Define duplet and octet rule.

(ii) Why ice floats over water?

(iii) Define metallic bond.

(iv) What is meant by standard atmospheric pressure?

(v) Define dynamic equilibrium.

(vi) What is diffusion? Explain with an example.

(vii) How molar solution is prepared?

(viii) Why are colloids quite stable?

Q4. (Marks 10) Write short answers to any Five (5) questions:

(i) Define oxidation number.

(ii) What is meant by electrolysis?

(iii) What are weak electrolytes? Give at least one example.

(iv) Define rusting. Write its important condition.

(v) What do you mean by 22 carat gold?

(vi) Write down chemical reaction of Mg with O_2 and N_2.

(vii) What is the trend of electropositivity in a group and a period?

(viii) Write any two physical characteristics of metals.

> Note: Attempt any TWO questions. (PART – II)

Q5.

(a) List five differences between compound and mixture.

(b) Describe four properties of canal rays.

Q6.

(a) Describe types of covalent bonds.

(b) Define evaporation. Explain factors affecting evaporation.

Q7.

(a) How can we prepare sodium hydroxide (NaOH) on commercial scale? Discuss its chemistry along with diagram.

(b) What is saturated solution and how it is prepared?